Counterfactual Selective Memory Repair (CSMR): Interdependent Long-Term LLM Agent Memory

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***Abstract*—Long-term memory allows LLM agents to retain information across conversations, but persistent memories can become inconsistent when later evidence changes the underlying state. The difficulty is not simply detecting that one memory is stale. A changed fact can make some other memories invalid, leave some conditionally valid, and have no effect on many semantically related memories. This creates a memory-repair scope problem: the agent must determine which existing records should be changed and which should be preserved. Recent research from 2023–2026 has addressed graph-based revision, knowledge-editing locality, long-term memory mechanisms, personal-memory selection, evidence-gated writing, persistent correction, and lifelong learning of LLM agents. These studies establish important individual mechanisms but do not directly settle the complete post-change decision over an already-existing external memory store. CSMR therefore investigates a discrete repair-set selection problem in which dependency is used to identify candidates, evidence sufficiency is evaluated for each candidate, and alternative repair sets are compared using downstream agent behavior while penalizing unnecessary changes. As implementation has not yet been completed, this paper presents the proposed method and expected evaluation rather than empirical CSMR results.**

I. Introduction

LLM agents increasingly operate across multiple interactions rather than producing one isolated response. In such systems, external or persistent memory can retain user facts, preferences, events, plans, previous decisions, summaries and other information for later retrieval. Recent IEEE work on lifelong LLM agents explicitly identifies memory as one of three core components, alongside perception and action, for continuous adaptation in dynamic environments. Persistent memory is also becoming a central part of long-term dialogue and personalized-agent architectures.

The benefit of long-term memory creates a maintenance problem. A memory can be correct when it is written and become incorrect after a later state change. Consider the following records:

- M_1: "The user owns a car."
- M_2: "The user drives to work by car."
- M_3: "The user spends money on fuel."
- M_4: "The user prefers driving."
- M_5: "The user likes Italian food."

Suppose the user later states, "I sold my car." M_1 is directly contradicted. M_2 is likely affected, while M_3 may require verification because the user could still use another vehicle. M_4 may remain true because selling a vehicle does not necessarily change a preference. M_5 is unrelated. A same-key update may therefore under-repair the memory store, whereas broad semantic propagation may over-repair it.

The central problem is consequently not only stale-memory detection. It is the choice of a repair set over already stored memories. The research question is:

> Given a state-changing observation and an existing persistent memory store, which memories should be repaired, what evidence is sufficient to justify each repair, and which repair set best restores downstream correctness while minimizing unnecessary modification?

A. Research Objectives

The proposed research aims to:

- represent new observations as structured state changes;
- identify direct conflict roots and a manageable candidate set;
- distinguish validity dependency from semantic similarity;
- evaluate evidence separately for each candidate memory;
- generate explicit repair alternatives instead of applying one propagation rule;
- use downstream question-answering or action correctness to compare repair alternatives; and
- preserve history through versioned, auditable memory updates.

B. Research Novelty

The proposal deliberately avoids broad claims such as being the first stale-memory detector, first memory graph, first evidence gate, or first selective-revision method. Those components already appear in recent work. The proposed novelty hypothesis is narrower: **post-change repair-set selection over an existing external memory store**. CSMR combines three decision stages: (1) learned cross-memory validity dependency for candidate construction, (2) per-memory evidence sufficiency for deciding whether a candidate should be changed, and (3) counterfactual comparison of complete repair sets using downstream behavior and explicit penalties for collateral changes and repair size.

The key distinction is therefore not that CSMR independently introduces dependency modeling, evidence assessment, or selective revision. Instead, it integrates these mechanisms around the specific decision of *which existing memories should*

actually be repaired after a state change. The proposed contribution is this joint repair-scope formulation and its evaluation through downstream behavior before committing an update.

II. Literature Review

The literature is restricted to work published from 2023 onward for the core review. It intentionally mixes IEEE papers and established journal/conference literature from SCIE/SCIE-oriented venues so that the research foundation is both recent and technically relevant.

A. Graph Revision, Knowledge Editing and Locality

SCIE – 2023: Graph-based mapping revision. Recent ontology-revision work constructs an integrated graph of ontology concepts and mappings, uses influence and impact relations, and localizes revision rather than forcing experts to reconsider every mapping. The study evaluates real-world ontology alignments and demonstrates that structural information can reduce manual revision effort. This is important to CSMR because it establishes that the relationships between knowledge items can determine which items deserve attention. The limitation is that the relationships are explicit in a symbolic ontology. The system does not have to infer hidden validity dependencies from free-form memory text. Therefore, its graph-based localization cannot directly answer whether a separately stored memory has become invalid because another memory changed. CSMR uses graph or local-link information only as part of candidate generation and does not treat graph connectivity itself as proof that a memory must be revised.

SCIE – 2023: Multi-edit knowledge editing. A divide-and-conquer knowledge-editing framework addresses interference when several knowledge edits are applied together. Instead of treating the edits as one monolithic change, the framework decomposes them into interacting groups and coordinates editing through parallel structures. The main contribution is the preservation of locality during multi-edit scenarios. However, the editable object is model knowledge represented in parameters rather than an external persistent memory database. The method also assumes that the requested edits are already known; it does not decide which existing external memory records should enter the edit scope. This leaves an earlier decision open for CSMR: before modifying memory, determine the records that are actually affected by a state change.

B. Knowledge Lifecycle and LLM-Agent Memory

SCIE/ESCI-status caveat – 2024: Knowledge-lifecycle survey. A knowledge-lifecycle survey organizes research around acquisition, representation, maintenance, editing and application. Its main value is conceptual: it separates knowledge maintenance from one-time acquisition and makes continual updating a distinct research stage. The limitation is that the survey does not define an operational post-change repair controller for external agent memory. It therefore serves as background rather than as the main technical basis of CSMR. Because the working literature review identified this paper as ESCI in the consulted indexing information, it should not be counted toward an institutional SCI/SCIE minimum unless explicitly permitted.

SCIE – 2025: LLM-agent memory survey. A recent survey organizes memory mechanisms for LLM agents by source, form, implementation and operations such as writing, management and reading, and distinguishes memory-level evaluation from downstream evaluation. This work establishes that memory is a first-class component of agent architecture and that memory quality must be evaluated beyond storage alone. Its limitation for CSMR is that it is a survey rather than a concrete post-change repair method. It does not define a learned validity- dependency classifier, a memory-specific evidence gate for existing records, or a repair-set objective. CSMR specializes this broad management problem into a state-change-triggered repair operation.

SCIE/SSCI – 2025: Personal-knowledge extraction with memory. A 2025 personal- knowledge extraction system uses task selection, memory selection, progressive extraction and knowledge re-identification so that historical memory can improve extraction from current data. Its contribution is to show how useful persistent personal information can be when interpreting new inputs. The limitation is the lifecycle position of the method: memory is used to improve today's extraction, but it does not primarily repair yesterday's stored knowledge after a later state change. Thus, a memory store can be useful for extraction while still containing stale facts. CSMR begins at this missing point: the memories already exist, and new evidence may invalidate some of them.

IEEE – 2025: Episodic-memory risks in AI agents. An IEEE Secure and Trustworthy Machine Learning paper argues that episodic memory will improve AI-agent capabilities but also introduce important safety and trust risks. It proposes principles for developing memory in ways that preserve monitoring, control and safety. The contribution for CSMR is the recognition that persistent memory is not only a capability problem but also a reliability and safety problem. Its limitation is that it does not provide a concrete repair algorithm or repair-set optimization. CSMR operationalizes one part of this reliability problem by making the update decision explicit and auditable.

C. Natural-Language and In-Context Knowledge Updating

SCIE/SSCI – 2026: Document-level knowledge editing. Recent document-level knowledge-editing research addresses the mismatch between clean factual triples and real information arriving in natural-language documents. It evaluates editing using dimensions such as edit success, locality, reasoning and cross-lingual transfer, and uses an Extract-then-Edit style pipeline. The key finding is that document-level updates are harder than clean-triple editing because extraction and editing interact. The limitation for CSMR is that the editable target remains model/document knowledge rather than a persistent external memory store. It does not decide which existing memory records are implicated by a state-changing observation. CSMR therefore treats natural-language evidence as input to

state-change extraction and candidate repair rather than as the final editable representation.

SCIE – 2026: Training-free in-context knowledge editing. DeCO addresses online knowledge change by using targeted information seeking, incremental reasoning and dynamic prefill guidance. Its setting is useful for CSMR because real corrections may arrive as natural-language evidence and require more than a single pass of reasoning. Its limitation is that the output is an edited reasoning/model state rather than a selected set of persistent memory records. There is no explicit cross-memory repair-set objective. CSMR can adopt targeted evidence processing while placing the repair-scope decision outside the reasoning editor.

D. Memory Admission, Validity and Downstream Consistency

SCIE – 2026: Scene-aware memory discrimination. A recent personal-memory system uses a Gating Unit Module and Cluster Prompting Module to decide which interaction information should remain in long-term memory according to scene and user intent. The study addresses memory noise and cost at the admission or retention stage. Its key contribution is to make memory selection explicit. Its limitation relative to CSMR is fundamental: admission asks whether information should be stored or retained when memory is being constructed, whereas CSMR starts with an already stored memory that was previously valid and later becomes affected by new evidence. This distinction prevents CSMR from claiming novelty in general memory selection.

SCIE – 2026: Belief consistency of LLM agents. Recent work on belief consistency uses generation, evaluation and attribution in an iterative loop to improve the consistency of multi-turn agent decisions. It demonstrates that inconsistent internal beliefs can affect downstream behavior and that evaluation should consider more than surface fluency. The limitation is that the method operates primarily at the belief and decision level. It does not identify which persistent memory records caused the inconsistency or select a minimal repair set. CSMR uses downstream behavior differently: as a signal for comparing alternative memory repairs before committing one.

SCIE – 2025: Persistent and transient refutation evaluation. Refutation benchmarking distinguishes immediate correction from persistence of that correction across later turns. This is useful because an agent may acknowledge new evidence once and still later return to an outdated state. The benchmark provides a way to expose long-horizon correction failure. However, it is an evaluation framework rather than a persistent-memory repair controller. It does not tell the system which records should be changed. CSMR can use persistent- refutation scenarios to evaluate whether a selected repair remains effective over later interactions.

E. Evidence, Consolidation and Lifelong Learning

IEEE – 2026: Evidence-gated memory writing. An IEEE ICIPAI paper introduces an Evidence-Gated Memory Writer that evaluates candidate memory entries using long-term utility, consistency with existing memory, and evidence sufficiency before allowing a write. It evaluates MemoryAgentBench and LoCoMo using three open-source LLMs and reports substantial reductions in memory pollution and improvements in task accuracy. The main limitation is timing: the gate controls memory creation, so it does not solve the case in which a memory was valid and well supported when written but becomes stale later. CSMR reuses the evidence-sufficiency principle at a different lifecycle point, namely post-change repair of existing records.

IEEE – 2026: Rapid episodic-merging memory. REMMEM addresses memory-management latency and fragmentation with a wake–sleep asynchronous architecture. Lightweight storage occurs during interaction, while background consolidation merges episodic memories and builds deeper semantic connections. The reported experiments show improvements in complex reasoning and reductions in token consumption and write latency. Its limitation is that consolidation and semantic validity are different objectives: memory can be well organized and still be stale. CSMR can be treated as a semantic repair layer that may run alongside or above such consolidation infrastructure.

IEEE – 2026: Lifelong learning of LLM-based agents. A 2026 IEEE TPAMI roadmap identifies perception, memory and action as three major modules for lifelong LLM-agent learning and emphasizes memory as the component that stores and retrieves evolving knowledge. It also discusses continuous adaptation, catastrophic forgetting, evaluation and long-term performance. The contribution for CSMR is the broader continual-learning context. Its limitation is that it is a roadmap/survey rather than a concrete repair-set algorithm. CSMR narrows this lifecycle challenge to one operational decision after a state change.

III. Research Gap

The reviewed literature shows that many individual components of memory maintenance are already established. Graph-based revision provides structural influence and localization. Knowledge-editing work studies locality and interference. LLM-agent memory research provides mechanisms for storing, retrieving, selecting and managing long-term information. Evidence-gated writing addresses whether new information should be admitted. Belief- consistency and refutation research demonstrate that memory or belief errors can influence later behavior. Lifelong-learning work places memory in the broader continual-adaptation loop.

However, these works address different lifecycle operations. A memory-admission method asks whether new information should enter memory. A stale detector asks whether a record is outdated. A validity system can revoke a record. A graph system can identify structurally related items. An evidence gate can reject weak information before it is stored. A downstream validator can check whether an action depends on correct information. None of these questions is identical to the following post-change decision:

After a state-changing observation, which existing external memories form the smallest justified repair set?

The distinction becomes important when semantic similarity, validity dependency and repair necessity diverge. Two records may be semantically related but independent. A dependent memory may not have sufficient evidence for immediate revision. Several valid repair candidates may also interact so that the best set is not the union of all high-scoring individual candidates.

A. Exact Research Gap

The research gap is therefore a joint post-change repair-scope problem over persistent external memory. The system needs to: (1) identify which stored memories are potentially validity-dependent on a changed state; (2) determine whether the available evidence is strong enough to change each candidate memory individually; and (3) choose the complete repair set that gives the best downstream correctness with minimal unnecessary modification.

IV. Proposed Method

CSMR is designed as a five-phase decision pipeline. The architecture separates candidate discovery from actual repair and delays permanent modification until alternative repair sets have been evaluated.

A. State-Change Extraction

Given a new observation e, a structured extraction process produces

$$\Delta = (entity, attribute, old, new, time, source, confidence).$$

For example, "I sold my car" becomes an ownership change with source and timestamp. The system then identifies direct conflict roots in the current memory store. No memory is modified at this stage.

B. Candidate Memory Construction

The system retrieves top-K potentially related memories using semantic embeddings and metadata filters for entity, attribute and time. It can expand one or two hops through local entity or provenance links. The result is a candidate set C. Candidate construction is deliberately permissive: it identifies memories that deserve analysis but does not assume that they are stale.

Candidate $\neq$ repair. A memory can enter C because it is potentially dependent on the changed state and still receive KEEP, VERIFY or CONDITIONAL if the evidence is insufficient for modification.

C. Dependency and Evidence Assessment

For each pair (m_i, m_j), dependency features can include semantic similarity, shared entity, shared attribute, temporal relation, causal cues, provenance relation and historical co-occurrence. A first implementation can use interpretable logistic regression:

$$P(D_{ij} = 1 \mid z_{ij}),$$

where D_{ij} denotes validity dependency and z_{ij} is the feature vector.

A separate evidence score evaluates whether the evidence is sufficient to modify candidate m_j:

$$S(E, m_j) = P(E \text{ is sufficient to change } m_j).$$

Possible evidence features include source reliability, freshness, directness, contradiction strength, corroboration, provenance quality and extraction confidence. Dependency therefore answers: "Should this memory enter the candidate set?" Evidence answers: "Is changing this candidate justified now?"

D. Counterfactual Repair-Set Selection

Instead of independently changing every candidate, CSMR generates a small family of repair alternatives. Typical alternatives include R_0: no change; R_1: repair only the direct root; R_2: repair the root plus high-confidence one-hop dependents; R_3: repair a deeper dependency chain; and mixed alternatives containing VERIFY or CONDITIONAL actions.

Each repair set is applied to a temporary copy of the memory store. Downstream questions or actions are then evaluated against each virtual state. This makes the counterfactual component concrete: the system asks what the agent would do if a particular repair set were applied, without permanently changing memory.

E. Minimal Commit and Validation

The best feasible repair set is selected, committed with versioning, and validated through a final downstream query or action. Previous memory versions are retained rather than permanently deleted. The system logs the evidence, dependency score, chosen repair action and resulting version transition.

F. Repair Actions

CSMR uses five explicit actions:

- **KEEP:** leave the memory unchanged.
- **VERIFY:** seek additional evidence before deciding.
- **CONDITIONAL:** retain the memory with a narrower context or validity condition.
- **REVISE:** update the memory content.
- **REVOKE:** deactivate the memory while retaining its history.

V. Mathematical Formulation

Let M be the current memory set, Δ the new state-changing evidence, R a candidate repair set and A a set of downstream questions or actions. The proposed objective is

$$R^* = \arg\max_R [Benefit(A \mid M \oplus R, \Delta) - \lambda_c Collateral(R) - \lambda_e Evide$$

The terms have the following meanings:

- **Benefit:** improvement in downstream answer or action correctness.
- **Collateral:** penalty for changing memories that should remain valid.
- **EvidenceCost:** cost of additional verification and model calls.

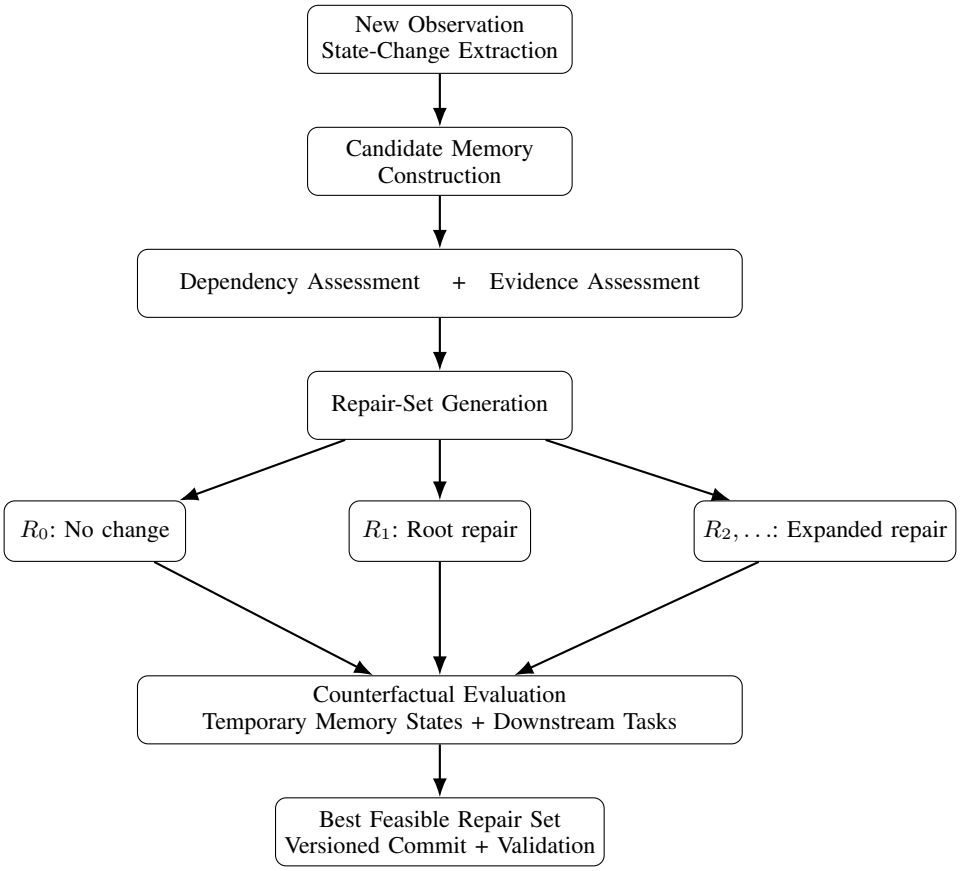


Fig. 1. CSMR decision pipeline. Candidate generation, evidence assessment, and repair-set selection are treated as separate decision stages before a versioned memory update is committed.

- $|R|$**:** number of changed memories, discouraging unnecessary broad repair.

The optimization turns the phrase "minimal selective repair" into a measurable decision rather than a descriptive label. The repair decision is additionally constrained by the candidate set and evidence sufficiency.

If m_i is outside the candidate set C, then its action must remain KEEP:

$$m_i \notin C \Rightarrow a_i = KEEP.$$

For modification actions, evidence must exceed a predefined sufficiency threshold τ_e:

$$a_i \in \{\text{REVISE}, \text{REVOKE}\} \Rightarrow S(E, m_i) \geq \tau_e.$$

Candidates with insufficient evidence may instead receive VERIFY or CONDITIONAL. These constraints prevent semantic similarity or graph connectivity alone from being treated as proof that a memory must be modified.

VI. Worked Example

Consider a persistent memory store containing the following records:

The new observation is:

"I sold my car."

TABLE I
INITIAL MEMORY STATE AND RELATIONSHIP TO THE STATE CHANGE.

ID	Content	Relation to change
M_1	The user owns a car.	Direct contradiction
M_2	The user drives to work by car.	Strong validity dependency
M_3	The user spends money on fuel.	Indirect dependency; not proven false
M_4	The user prefers driving.	Semantically related but not logically implied
M_5	The user likes Italian food.	Unrelated

A last-write-wins method may change only M_1, producing under-repair because M_2 can still cause the agent to answer as though the user drives the sold vehicle. A broad semantic-propagation method may change M_1 through M_4, producing over-repair because the observation does not prove that the user stopped preferring driving.

CSMR first extracts the state change as an ownership transition and identifies M_1 as the direct conflict root. Candidate construction then expands to M_2 and M_3 because they share relevant entities or validity dependencies. M_4 may be retrieved because it is semantically related, but dependency analysis should assign it a lower validity-dependency probability. M_5

TABLE II
CSMR REASONING FOR THE WORKED EXAMPLE.

ID	Dependency	Evidence assessment	Action
M_1	Direct	Strong contradiction; high confidence	REVOKE/REVISE
M_2	Strong	Ownership change directly affects stated vehicle use	REVISE
M_3	Indirect	Selling one vehicle does not prove fuel spending stopped	VERIFY
M_4	Weak / independent	Preference is not contradicted by ownership change	KEEP/CONDITIONAL
M_5	None	No relevance to changed state	KEEP

should not enter the candidate set.

CSMR then constructs alternative repair sets:

- R_0: no change;
- R_1: REVOKE/REVISE M_1 only;
- R_2: repair M_1 and M_2;
- R_3: repair M_1, M_2 and M_3;
- R_4: repair M_1–M_4.

The alternatives are evaluated on downstream tasks such as "How does the user commute to work?" and "What vehicle does the user currently own?" R_0 leaves contradictory information available and therefore can produce incorrect answers. R_1 removes the direct contradiction but may leave M_2 active, allowing the agent to infer that the user still drives the sold car. R_2 addresses both the direct fact and its strongest dependent memory. R_3 may be unnecessarily aggressive because the observation does not establish that all fuel spending has stopped. R_4 introduces an even stronger collateral-change risk because the user's preference for driving is not logically invalidated.

Under the proposed objective, R_2 is therefore a plausible preferred repair set: it improves downstream correctness while avoiding unsupported changes to M_3 and M_4. M_3 can remain under VERIFY, M_4 can remain KEEP or CONDITIONAL, and M_5 remains untouched. The example demonstrates the central CSMR principle: **being related to a changed fact is not sufficient for repair; repair requires both validity dependency and adequate evidence.**

VII. Expected Results and Discussion

The project is currently pre-implementation. Therefore, no empirical CSMR accuracy, F1 score, benchmark result or performance improvement is claimed. The following hypotheses define the expected outcomes to be tested.

- **H1 – Dependency:** Learned validity dependency will identify cross-key and multi-hop affected memories better than same-key replacement or semantic similarity alone.
- **H2 – Evidence:** Memory-specific evidence sufficiency will reduce unnecessary revisions under weak or ambiguous evidence.
- **H3 – Counterfactual selection:** Counterfactual repair-set comparison will improve downstream question-answering and action correctness compared with dependency-only or broad propagation strategies.
- **H4 – Minimality:** Explicit minimality and collateral-change penalties will reduce unnecessary modification while preserving repair recall.

These hypotheses also make the expected trade-off explicit. CSMR may require more computation than a single-pass update because it evaluates alternative repair sets. The expected benefit is therefore not minimum latency alone, but improved correctness and lower collateral modification for persistent memory.

VIII. Planned Evaluation

Benchmark construction. The evaluation will use controlled dynamic-memory scenarios. Each scenario will contain an initial persistent memory set, a later state-changing observation, ground-truth repair actions for affected memories, and downstream questions or actions whose correctness depends on the updated memory state. Scenarios will cover direct contradictions, cross-key dependencies, two-hop and three-hop dependencies, false semantic similarity, temporal/context-dependent validity, conflicting sources, weak evidence, derived summaries and plans, memory poisoning or malicious observations, and persistent refutation across multiple turns.

To reduce the risk of template memorization, the benchmark should be divided into training, validation and held-out test scenarios, with scenario templates or entity combinations in the test set kept distinct from those used for training. A smaller manually verified subset can be used first to validate benchmark construction before scaling it.

Baselines. Planned baselines include append-only memory, recency-only updating, same-key last-write-wins, semantic-similarity propagation, fixed graph propagation, dependency-only revision, evidence-only revision, and recent validity/stale-memory approaches such as TEPA- or STALE/CUPMem-style mechanisms. Every method will receive the same initial memory state, state-changing observation and downstream evaluation tasks so that differences can be attributed to the repair strategy rather than different inputs.

Metrics. Dependency precision/recall/F1 will measure the quality of validity- dependency prediction. Repair precision will measure the fraction of changed memories that truly required modification, while repair recall will measure the fraction of memories requiring repair that were correctly addressed. Over-invalidation will measure valid memories that were unnecessarily modified, and under-repair will measure affected memories that remain incorrect. Downstream QA accuracy and action success will measure the effect of the repaired memory state on agent behavior. Persistence of correction will test whether the corrected state remains effective across later turns. Evidence cost and latency will quantify the computational overhead of selective repair.

Stress testing and ablation. The evaluation will separately test weak evidence, conflicting sources, temporary state changes, multi-hop dependencies, semantically similar but independent memories, and malicious observations. An ablation analysis will compare full CSMR against variants without dependency assessment, evidence assessment, counterfactual

repair-set evaluation, and minimality penalties. This will test whether each major component contributes to selective repair rather than simply increasing overall reasoning cost.

Error analysis. Incorrect CSMR decisions will be categorized into state-change extraction errors, candidate-retrieval failures, dependency-classification errors, evidence-assessment errors, counterfactual-evaluation errors, and genuinely ambiguous cases. This analysis will identify the conditions under which selective repair remains unreliable and will distinguish limitations of individual components from limitations of the overall repair-set formulation.

IX. Conclusion

The post-2023 literature shows that important ingredients of long-term memory maintenance are already being investigated: graph-based revision, knowledge-editing locality, memory management, memory admission, evidence gating, continual learning and downstream consistency. The research problem should therefore not be stated as "stale memory has not been solved."

The narrower CSMR question is: after a state-changing observation, how can an agent choose the smallest evidence-justified repair set over already-stored external memories? The proposed method separates candidate dependency from evidence sufficiency and from the final joint repair decision. Alternative repair sets are tested counterfactually before versioned commitment, with downstream behavior included in the objective.

The proposed evaluation is designed to determine whether this formulation improves repair precision, repair recall, downstream correctness and persistence while reducing under-repair, over-invalidation and unnecessary repair cost. The final empirical contribution will depend on implementation, controlled evaluation, baseline comparisons, ablation studies and a final closest-work audit immediately before submission.

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